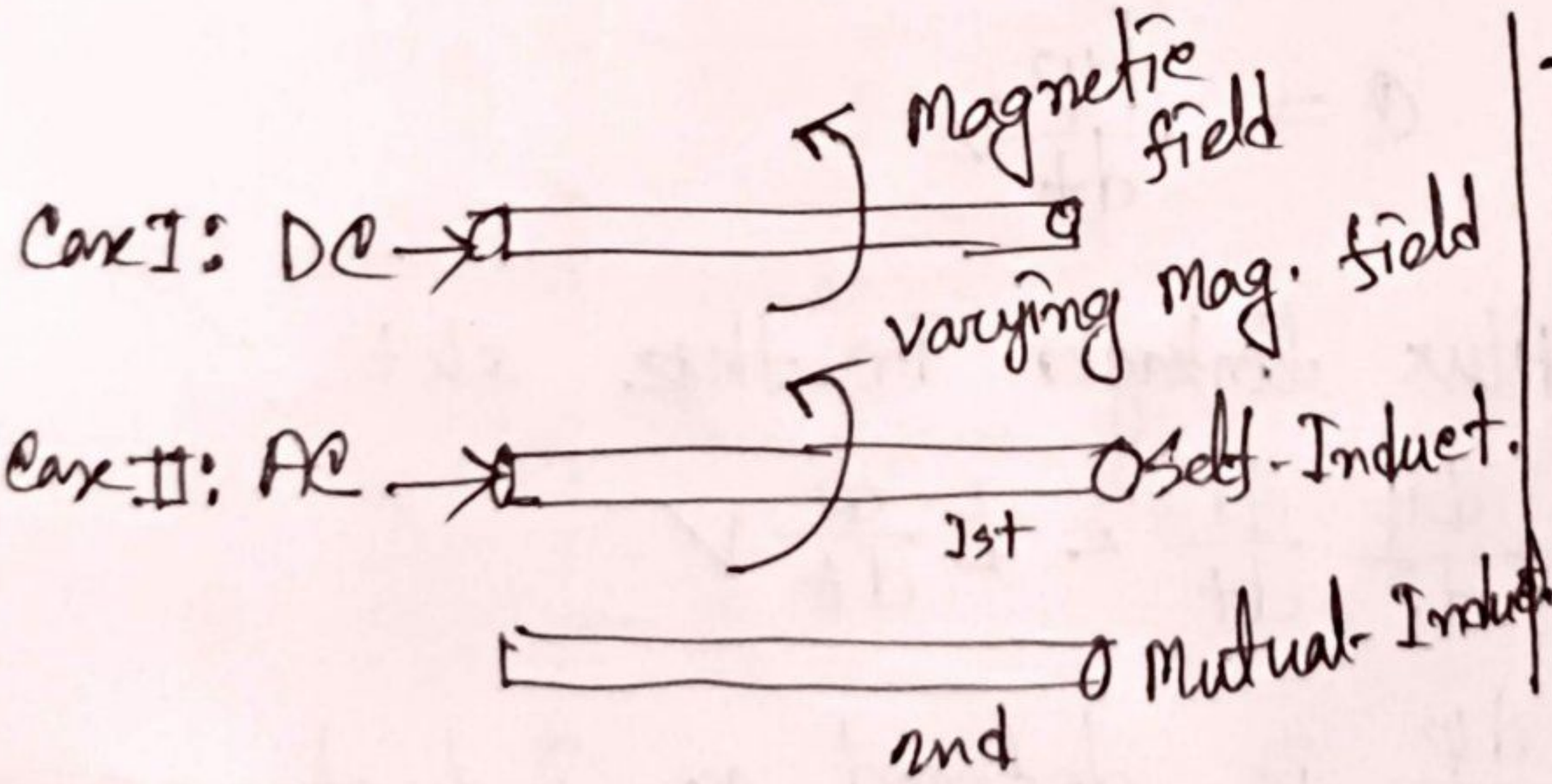
Chap: 02

Inductance and Resistance of Transmission Lines:

Performance of Transmission line as an element of power system depends on 4 parameters:

i) Inductance ii) Capacitance

iii) Resistance iv) Conductance.



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AC currents passed through the 1st conductor, varying magnetic field will produce. Applying to Faraday's laws, varying magnetic field will produce EMF

↓

Flux linkage with conductor charges

↓

Inductance

Art 2.2

Voltage induced in a ckt,

$$e = \frac{d\psi}{dt} V$$

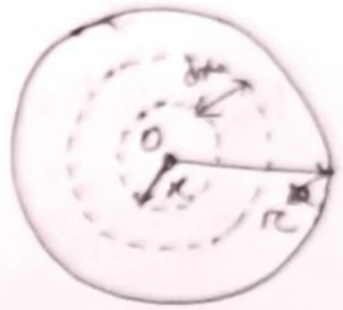
Here, ψ = Flux linkages in the ckt

$$e = \frac{d\psi}{di}\frac{di}{dt} = L\frac{di}{dt} V$$

Here, $L = \frac{d\psi}{di}$ is defined as Inductance

Art 2.3:

Flux linkage due to Internal flux:



At distance x, hop thickness dx, conductor radius r.

Assume, some current density. H_x = Magnetic flux Intensity.

Ampere's law,

$$mmf = \oint H_x dl = I_x \qquad H_x \to \text{Const.}$$

$$\Rightarrow H_x(2\pi x) = I_x$$

$$\Rightarrow H_x = \frac{I_x}{2\pi x} \quad \text{--- (1)}$$

As ~~constant~~ current density is uniformed

$$J = J_x \Rightarrow \frac{I}{A} = \frac{I_x}{A_x}$$

$$\Rightarrow \frac{I}{\pi r^2} = \frac{I_x}{\pi x^2}$$

$$\Rightarrow \frac{I_x}{I} = \left(\frac{x}{r}\right)^2 \quad \text{--- (n)}$$

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① → $H_x = \frac{x^2}{r^2} I \frac{1}{2\pi x}$

Flux density at point x.

$$B_x = \mu H_x$$
$$= \mu \frac{x^2}{r^2} I \frac{1}{2\pi x}$$
$$= \frac{\mu x I}{2\pi r^2}$$

Flux linkage $d\varphi$ with current per meter length of conductor

$$d\lambda = \frac{I_x}{I} d\varphi$$
$$= \left(\frac{x}{r}\right)^2 \frac{\mu x I}{2\pi r^2} dx$$
$$= \frac{\mu x^3 I}{2\pi r^4} dx \text{ WbT/m}$$

$\phi = A \cdot B$

$d\phi = (L \cdot dx) B_x$

$= l_m \times dx \cdot B_x$

$= dx \frac{\mu x I}{2\pi r^2}$

Total internal flux linkage.

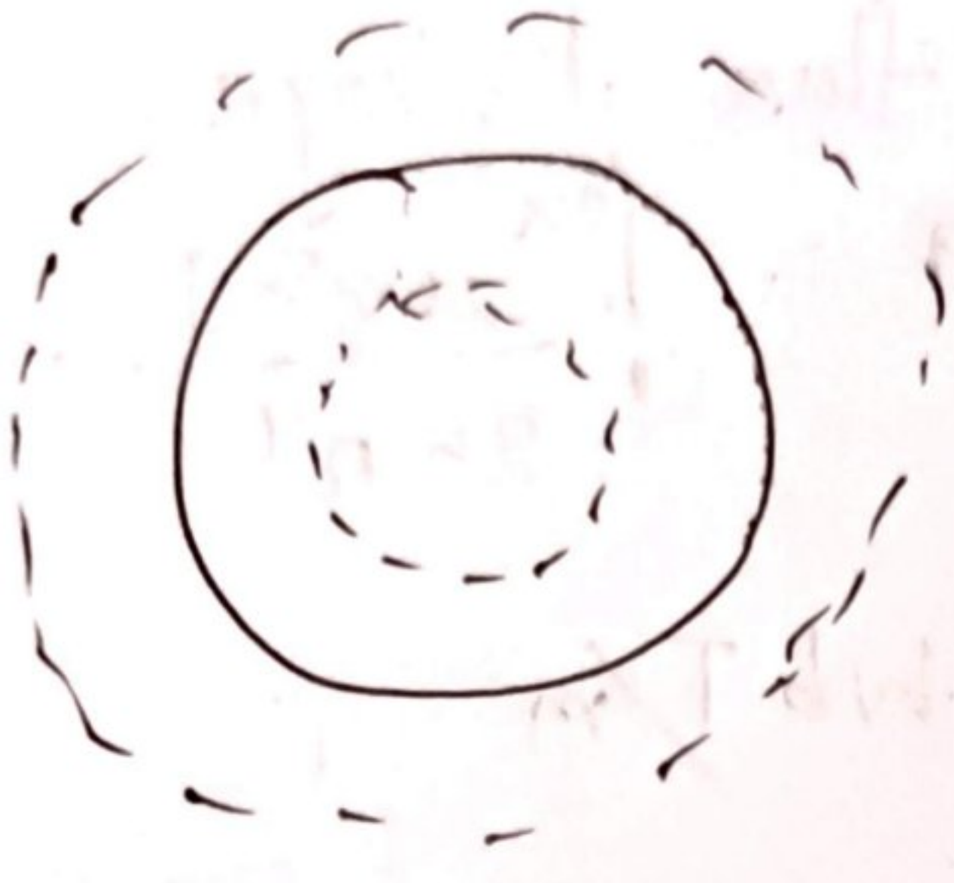
$$\lambda_{int} = \int d\lambda = \int_0^{2\pi} \frac{\mu x^3 I}{2\pi r^4} \cdot dx$$
$$= \frac{\mu I}{8\pi} \text{ WbT/m}$$
$$= \frac{\mu_0 \mu_r I}{8\pi} = \frac{4\pi \times 10^{-7} \mu_r I}{8\pi}$$
$$= \frac{1}{2\pi} \mu_r I \times 10^{-2} = \frac{1}{2} I \times 10^{-7} \quad (\text{Non-magnetic } \mu_r = 1)$$

Internal inductance per meter length,

$$L_{in} = \frac{\lambda}{I} = \frac{1}{2} I \times 10^{-7}$$
$$= \frac{1}{2} \times 10^{-7} \text{ H/m}$$

Without considering proximity effect, skin effect.

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$$L_{total} = L_{internal} + L_{external}$$

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